A curvature dependent lemniscate method for terminal gradients

Abstract

We give a horizon dependent first-order method for the terminal gradient of a smooth strongly convex function. Its coefficients are determined by a unique scalar boundary-value problem that deforms the known convex lemniscate recurrence. The method uses two vector accumulators and N gradient evaluations to construct its output. An explicit rank-two interpolation certificate proves a gradient-norm bound with factor $\tau_N(\mu/L)$. A fixed quadratic attains this bound, so the factor is the exact worst-case value for the constructed method. It decreases strictly with the curvature ratio and reduces to the convex lemniscate factor when $\mu = 0$. On an explicit positive-curvature interval, it improves three specified bounds obtained from convex class inclusion and existing ITEM variants. No minimax optimality among competing methods is claimed.

Keywords: smooth strongly convex optimization; gradient norm; lemniscate acceleration; performance estimation

2020 Mathematics Subject Classification. Primary 65K10; Secondary 90C25, 90C60.

1 Introduction

Terminal-gradient guarantees for first-order methods depend on both the objective class and the quantity used to normalize the starting point. For smooth convex functions, Kim, Ryu, and Das Gupta [3] give lemniscate acceleration, whose gradient-norm factor is defined by a scalar path. Grimmer, Jo, and Park [5] establish its optimality and uniqueness among fixed-step first-order methods in that convex setting when the dimension is at least $N + 2$. These results concern the convex class, without a positive lower curvature bound.

Shu and Wang [4, Example 2] report a numerical distance-to-gradient method for a smooth strongly convex class and point to an analytic description as a further direction. Their certificate-level H-duality identifies a self-dual sparse pattern, but does not supply an all-horizon positive-curvature recurrence or rate formula. The distance-to-distance method ITEM [2], its gradient-to-gradient dual [4], and the function-gap method ITEM-f [3] control different pairs of initial and terminal criteria.

We construct a curvature dependent deformation of the lemniscate path and an explicit two-accumulator method. The main theorem gives a dimension-independent terminal-gradient bound for every finite horizon and every $0 \le \mu < L$. Its proof starts from the actual recurrence and verifies a final rank-two Gram certificate. The convex path, interpolation inequalities, sparse certificate structure, and H-duality framework are established antecedents [1, 3–5]. The positive-curvature scalar path and its connection to the method and certificate are the content proved here.

The bound is attained by a two-dimensional quadratic started in its L-eigendirection. This establishes the exact worst-case factor of this method, while leaving its minimax optimality among competing methods open. The comparison results concern three specified guaranteed coefficients on a bounded curvature interval. The coefficients depend on the prescribed horizon, and the method requires the smoothness and curvature parameters. No initial function gap or knowledge of the minimizer is used.

2 The scalar path and the method

Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable, μ-strongly convex, and have L-Lipschitz gradient, where $L > 0$ and $0 \le \mu < L$. Suppose f has a minimizer y_*. Write

$$\kappa = \frac{\mu}{L}, \qquad q = \frac{\kappa}{1-\kappa}. \tag{1}$$

The parameter q is the transformed curvature, rather than μ/L itself.

For an integer $N \ge 1$, define $\tau = \tau_N(\kappa)$ and a decreasing path by

$$\begin{aligned} 1 = \rho_0 > \rho_1 > \cdots > \rho_N > \rho_{N+1} = 0, \\ (\rho_i - \rho_{i+1})^2 = \rho_i(1-\rho_{i+1})\big[\kappa(1-\rho_{i+1}) + (1-\kappa)\tau(1+\rho_{i+1})\big], \quad 0 \le i \le N. \end{aligned} \tag{2}$$

Existence and uniqueness are proved in Section 3. Set

$$\begin{aligned} T = \frac{1}{\tau+q}, \qquad \gamma = \frac{\tau+q}{\tau}, \quad p_i = 1-\rho_{i+1}, \\ C_i = \frac{\gamma p_i^2}{2(1-p_i)}, \quad a_i = p_i + C_i, \quad 0 \le i < N, \end{aligned} \tag{3}$$

and put $p_{-1} = C_{-1} = 0$, $\Delta p_i = p_i - p_{i-1}$ and $\Delta C_i = C_i - C_{i-1}$. All denominators in (3) are positive.

Starting from arbitrary y_0, initialize $A = B = 0$. For $i = 0, \dots, N-1$, perform

$$\begin{aligned} &y_i = y_0 + \frac{(1+q)T}{a_i}\big[(\gamma p_i + C_i)A - (1-p_i)B\big] \qquad (i > 0), \\ &s_i = \frac{1}{L}\nabla f(y_i), && w_i = q(y_i - y_0) - (1+q)s_i, \\ &A \leftarrow A + \Delta p_i w_i, && B \leftarrow B + \Delta C_i w_i. \end{aligned} \tag{4}$$

The value y_0 at $i = 0$ is the given starting point. After these N evaluations, output

$$y_N = y_0 + (1+q)TA. \tag{5}$$

The accumulator updates both use the displayed w_i. Only two vector accumulators, the starting point and the current query are needed. The scalar coefficients are determined before any oracle observations. Evaluating the terminal gradient to measure the criterion would require an additional oracle call; it is not needed to construct (5).

Theorem 1. *For every $N \geq 1$ and $0 \leq \kappa < 1$, problem (2) has a unique solution with $0 < \tau_N(\kappa) < 1$. The method (4)–(5) satisfies*

$$\|\nabla f(y_N)\| \leq L\tau_N(\kappa)\,\|y_0 - y_*\|\,. \tag{6}$$

For fixed $N \geq 1$, τ_N is continuous and strictly decreasing on $[0,1)$. It also decreases strictly with N, and

$$\tau_1(\kappa) = \frac{1-\kappa}{1+\sqrt{1+(1-\kappa)^2}}. \tag{7}$$

At $\kappa = 0$, one has $\tau_N(0) = 1/\Omega_N$, where Ω_N is the convex lemniscate constant at horizon N in [3, Lemma 1 and Theorem 1]. For $N = 0$, setting $\tau_0 = 1$ and returning y_0 gives (6).

Section 4 proves (6) by an exact certificate. Proposition 4.1 shows that equality is attained, so τ_N is the exact worst-case factor of the constructed method. Section 5 establishes the specified benchmark improvements.

3 The scalar boundary-value problem

Fix $0 \leq \kappa < 1$ and a prospective $\tau \in [0,1]$. To avoid ambiguity between the two curvature normalizations, write

$$r = (1-\kappa)\tau, \qquad c = \kappa + r, \qquad b = r - \kappa.$$

The edge polynomial in (2) is

$$\begin{aligned} F(x,y) &= (x-y)^2 - x(1-y)(c+by) \\ &= (1+bx)y^2 - 2(1-\kappa)xy + x(x-c). \end{aligned} \tag{8}$$

Its quadratic coefficient is at least $1-\kappa > 0$ for $0 \leq x \leq 1$.

Lemma 1. *There is a jointly continuous map $\Phi_{\kappa,\tau} : [0,1] \to [0,1]$ obtained by taking the smaller root of (8) and replacing a negative root by zero. For $0 < \tau < 1$, it vanishes on $[0,c]$. For $c < x < 1$, its value is the unique root in $(0,x)$, and*

$$\Phi_{\kappa,\tau}(1) = h(\tau) := \frac{1-\tau}{1+\tau}. \tag{9}$$

The map is nondecreasing in x and nonincreasing in both τ and κ.

Proof. Since $F(x,x) = -x(1-x)(c+bx) \leq 0$, the discriminant is nonnegative. The explicit map is

$$\Phi_{\kappa,\tau}(x) = \max\left\{0,\, \frac{(1-\kappa)x - \sqrt{(1-\kappa)^2x^2 - (1+bx)x(x-c)}}{1+bx}\right\}. \tag{10}$$

This proves joint continuity, including $\tau = 0$ and the clamping threshold. For $c < x < 1$, $F(x, 0) > 0$ and $F(x, x) < 0$, so exactly one root lies in $(0, x)$; the other is greater than x. At $x = 1$,

$$F(1, y) = (1 - \kappa)(1 - y)\big[(1 - \tau) - (1 + \tau)y\big],$$

which gives (9). The root $y = 1$ does not give strict progress.

On a positive strict-progress branch, $F_y < 0$. Using $F(x, y) = 0$ gives

$$F_x = (x - y)(1 + y/x) > 0, \qquad F_\tau = -(1 - \kappa)x(1 - y)(1 + y) < 0.$$

Implicit differentiation therefore proves increase in x and decrease in τ on this branch. Continuity extends these monotonicities over the zero part. In particular, every root satisfies $y \le h(\tau)$. At fixed τ,

$$F_\kappa = -x(1 - y)\big[1 - \tau - (1 + \tau)y\big] \le 0$$

on the positive branch, so the root is nonincreasing in κ there as well. Continuity and the clamped value extend the assertion to the boundary. The cases $\tau = 0, 1$ follow by continuity. ∎

Proof of the scalar assertions in Theorem 1. Define $x_0(\tau) = 1$ and $x_{j+1}(\tau) = \Phi_{\kappa,\tau}(x_j(\tau))$. For fixed $N \ge 1$ the function

$$x_N(\tau) - \big[\kappa + (1 - \kappa)\tau\big]$$

is continuous and strictly decreasing: its first term is nonincreasing and its second term strictly increasing. At $\tau = 0$, formula (10) gives $x_j(0) = 1$ for every finite j, so the difference is $1 - \kappa > 0$. At $\tau = 1$, the first iterate is zero and the difference is -1. There is consequently one root $\tau_N \in (0, 1)$.

At this root $x_N = \kappa + (1 - \kappa)\tau_N > 0$. None of the preceding iterates was clamped, and all preceding steps are strictly decreasing. The last step from x_N to zero satisfies the edge equation exactly. These values form the required path. Conversely, every strictly decreasing solution must use the smaller root at every nonterminal step. Its terminal equation forces $\rho_N = \kappa + (1 - \kappa)\tau$, so the shooting root and the path are unique. In particular, a prematurely clamped zero is never used as an equality solution.

Joint continuity of the shooting map and the strict crossing imply continuity of τ_N in κ: two parameters bracketing the unique root continue to bracket it under a sufficiently small change of κ. At fixed $0 < \tau < 1$, x_N is nonincreasing in κ, whereas $\kappa + (1 - \kappa)\tau$ is strictly increasing. The unique crossing therefore decreases strictly with κ.

At $\tau = \tau_N$, the next shooting iterate is zero. The crossing function for $N + 1$ is then negative, which proves $\tau_{N+1} < \tau_N$. For $N = 1$, the equation

$$\frac{1 - \tau}{1 + \tau} = \kappa + (1 - \kappa)\tau$$

gives (7). Finally, when $\kappa = 0$, equation (2) becomes

$$\tau^{-1}(\rho_i - \rho_{i+1})^2 = \rho_i(1 - \rho_{i+1}^2).$$

This is exactly the defining normalization of Ω_N in [3, Lemma 1]; uniqueness identifies $\tau_N(0) = 1/\Omega_N$. ∎

Lemma 2. *For the unique path, let*

$$J(x) = \frac{c(1-x)}{c+bx}. \tag{11}$$

Then $\rho_{N+1-i} = J(\rho_i)$, and the coefficients in (3) satisfy

$$a_i = \frac{p_i}{p_{N-1-i}}, \qquad a_i a_{N-1-i} = 1. \tag{12}$$

In particular, C_i and p_i are positive and strictly increasing.

Proof. The denominator $c + bx = \kappa(1-x) + r(1+x)$ is positive on $[0,1]$. Direct algebra gives $J(J(x)) = x$, $J(0) = 1$, $J(1) = 0$, and

$$F(J(y), J(x)) = \frac{4c^2r^2}{(c+bx)^2(c+by)^2} F(x,y). \tag{13}$$

The reversed-reflected path is therefore another strictly decreasing solution with the same endpoints and parameter. Uniqueness proves the symmetry. In the p_i variables it reads

$$p_{N-1-i} = \frac{2\tau(1-p_i)}{2\tau + (q-\tau)p_i},$$

which gives (12) after substituting (3). Positivity and increase of C_i also follow directly from $C_i = \gamma p_i^2/[2(1-p_i)]$ and $0 < p_0 < \cdots < p_{N-1} < 1$. ∎

The shooting construction gives a one-dimensional way to determine the coefficients. It does not require a finite-horizon matrix optimization. The endpoint $\rho_0 = 1$ must use (9), and the terminal equality must be enforced; repeated endpoint values would change the definition.

4 An exact certificate for the actual recurrence

We normalize $L = 1$ and translate the minimizer to the origin, subtracting $f(y_*)$ from f. The algorithm is translation invariant, since its accumulator inputs use $y_i - y_0$. The normalized objective is κ-strongly convex and 1-smooth.

4.1 Interpolation and the lifted inequality

For each point y_i, write

$$s_i = \nabla f(y_i), \qquad x_i = y_i - s_i, \qquad g_i = s_i - q x_i, \qquad F_i = f(y_i) - \tfrac{1}{2}\|s_i\|^2 - \tfrac{q}{2}\|x_i\|^2 . \tag{14}$$

Include the optimizer index $*$, with $x_* = g_* = F_* = 0$. The smooth strongly convex interpolation inequality [1] gives

$$\begin{aligned} F_i - F_j - \langle g_j, x_i - x_j\rangle = f(y_i) - f(y_j) - \langle s_j, y_i - y_j\rangle - \tfrac{1}{2}\|s_i - s_j\|^2 \\ - \tfrac{q}{2}\|y_i - y_j - s_i + s_j\|^2 \geq 0. \end{aligned} \tag{15}$$

This is the transformed interpolation used in [4].

Regard x, g, s as matrices whose rows are the corresponding vectors. Let Λ have nonnegative off-diagonal entries and nonpositive row and column sums. Put edge weights Λ_{ij} on $i \to j$ for $i \neq j$, add $i \to *$ with weight $-(\Lambda\mathbf{1})_i$, and add $* \to i$ with weight $-(\Lambda^T\mathbf{1})_i$. The total incoming and outgoing weights at each vertex agree. Summing (15) in the orientation $F_j - F_i - \langle g_i, x_j - x_i\rangle \geq 0$ therefore cancels all function values and proves

$$\operatorname{tr}(g^T \Lambda x) \leq 0. \tag{16}$$

Let S be the $(N+1)\times(N+1)$ lower-triangular cumulative-sum matrix. A strictly lower-triangular method matrix $\widetilde{H}$ represents

$$y_n = y_{n-1} - \sum_{j<n} \widetilde{H}_{nj} s_j.$$

Set

$$\begin{aligned} V &= (I + S\widetilde{H})\big[(1+q)I + qS\widetilde{H}\big]^{-1}, \\ v &= (I - qV)\mathbf{1}, \qquad w = (I - qV)^T e_N, \qquad \zeta = q e_N^T v. \end{aligned} \tag{17}$$

The inverse exists by triangularity. Substituting $y = \mathbf{1}y_0 - S\widetilde{H}s$ into (14) yields

$$x = -Vg + v y_0, \qquad s_N = w^T g + \zeta y_0. \tag{18}$$

Lemma 3. *Suppose Λ has the stated signs and*

$$\mathcal{L} = \begin{pmatrix} -\Lambda V - V^T\Lambda^T & \Lambda v & w \\ v^T\Lambda^T & \tau & \zeta \\ w^T & \zeta & \tau \end{pmatrix} \succeq 0, \qquad \tau > 0. \tag{19}$$

Then $\|s_N\| \leq \tau \|y_0\|$.

Proof. The quadratic form of (19), divided by two and evaluated on (g, y_0, ϵ), is

$$\operatorname{tr}(g^T\Lambda x) + \frac{\tau}{2}\|y_0\|^2 + \langle s_N, \epsilon\rangle + \frac{\tau}{2}\|\epsilon\|^2 .$$

It is nonnegative for every auxiliary vector ϵ. Minimizing in ϵ and using (16) gives

$$0 \leq \operatorname{tr}(g^T\Lambda x) + \frac{\tau}{2}\|y_0\|^2 - \frac{1}{2\tau}\|s_N\|^2 \leq \frac{\tau}{2}\|y_0\|^2 - \frac{1}{2\tau}\|s_N\|^2 .$$

This proves the assertion. The argument is the distance-to-gradient lift of [4, Lemma 9], with its norm-rate normalization made explicit. ∎

4.2 Recovering the transformed matrix from the accumulators

Let $X_i = y_i - y_0$ and denote the accumulator input by $u_i = qX_i - (1+q)s_i$ in this subsection, to distinguish it from the vector w in (17). Immediately before query i, the accumulators in (4) are

$$A = \sum_{j<i} \Delta p_j u_j, \qquad B = \sum_{j<i} \Delta C_j u_j.$$

Thus the actual algorithm is $X = Ku$, where K is strictly lower triangular and

$$\begin{aligned} K_{ij} &= \frac{(1+q)T}{a_i}\big[(\gamma p_i + C_i)\Delta p_j - (1-p_i)\Delta C_j\big] \quad (j<i<N), \\ K_{Nj} &= (1+q)T\Delta p_j \qquad\qquad\qquad\qquad\qquad\quad (j<N). \end{aligned} \tag{20}$$

Since $u = qX - (1+q)s$,

$$X = -B_{\mathrm{met}}s, \qquad B_{\mathrm{met}} = (1+q)(I-qK)^{-1}K, \qquad \widetilde{H} = S^{-1}B_{\mathrm{met}}.$$

These matrices are strictly lower triangular. Substitution into (17) simplifies the transformed matrix to

$$V = d(I+K), \qquad d = \frac{1}{1+q}. \tag{21}$$

All inverses here are of triangular matrices with nonzero diagonal. In particular, the representation does not require the unqueried gradient s_N to execute the algorithm.

Put $Z = VS^T$ and let t_i be the sum of row i of V. For $j<i<N$, formula (20) gives

$$Z_{ij} = \frac{T}{a_i}\big[(\gamma p_i + C_i)p_j - (1-p_i)C_j\big]. \tag{22}$$

For $j \ge i$, $Z_{ij} = t_i$. We need the exact row sums

$$t_i = \frac{TC_i}{a_i} \quad (i<N), \qquad Z_{Nj} = Tp_j \quad (j<N), \qquad t_N = T. \tag{23}$$

To verify the first identity directly, set $x = \rho_i$, $y = \rho_{i+1}$, $p = 1-y$, $h = 1-x$, and write $C(p) = \gamma p^2/[2(1-p)]$, $a(p) = p + C(p)$. Algebra gives

$$\begin{aligned} c\,a(p) + (\gamma p + C(p))h - (1-p)C(h) - C(p) \\ = -\frac{\gamma}{2xy}F(x,y). \end{aligned} \tag{24}$$

Here $x, y > 0$ because $i < N$. The path makes the right side zero, and $d/T = c$. Since the row sum obtained from (21) is

$$t_i = d + \frac{T}{a_i}\big[(\gamma p_i + C_i)p_{i-1} - (1-p_i)C_{i-1}\big],$$

identity (24) proves $a_i t_i = TC_i$, including $i = 0$. For the last row, $Z_{Nj} = Tp_j$ and $t_N = d + Tp_{N-1} = T$, because the last path equation gives

$$\rho_N = \kappa + (1-\kappa)\tau = \frac{q+\tau}{1+q} = \frac{d}{T}. \tag{25}$$

4.3 The rank-two Gram identity

Define a sparse matrix by

$$\begin{gathered}\Lambda_{ii} = -a_i, \qquad \Lambda_{i+1,i} = a_i, \qquad \Lambda_{iN} = \Delta C_i \quad (0 \le i < N),\\ \Lambda_{NN} = -(1 + C_{N-1}),\end{gathered} \tag{26}$$

with all remaining entries zero. Its off-diagonal entries are nonnegative. Its row sums are $-p_0, -\Delta p_1, \dots, -\Delta p_{N-1}, -(1 - p_{N-1})$, and its column sums vanish except for the last one, which equals -1. Thus (16) applies.

The matrix has the sparse pattern motivating the distance-to-gradient example in [4]. Its convex optimal specialization is explicit in [5, Section 5.3 and Appendix E.3]. The following proof uses its signs and exact entries directly.

Let $F_\Lambda = S\Lambda$ and $C = (C_0, \dots, C_{N-1})^T$. Then

$$F_\Lambda = \begin{pmatrix} -\operatorname{diag}(a_0, \dots, a_{N-1}) & C \\ 0^T & -1 \end{pmatrix}.$$

Congruence of (19) by $\operatorname{diag}(S, 1, 1)$ gives a matrix M whose leading block is $-F_\Lambda Z - Z^T F_\Lambda^T$. Formulas (22)–(23) yield, for $i, j < N$,

$$M_{ij} = \frac{p_i p_j}{\tau}, \quad M_{iN} = T p_i, \quad M_{iD} = -p_i, \quad M_{iG} = -qT p_i. \tag{27}$$

Here D, G name the last two lifted indices, not algorithmic iterates. On the diagonal, the first formula is the identity $2TC_i(1 - p_i) = p_i^2/\tau$; off the diagonal it follows by substitution in (22). The three anchor indices have principal block

$$M_{N,D,G} = \begin{pmatrix} 2T & -\tau T & \tau T \\ -\tau T & \tau & q\tau T \\ \tau T & q\tau T & \tau \end{pmatrix}. \tag{28}$$

For example, the transformed last column is $Sw = e_N - q(Z_{N0}, \dots, Z_{NN})^T$; the term $Se_N = e_N$ fixes the endpoint normalization.

Define vectors of length $N + 3$ by

$$\begin{aligned}\xi &= (p_0, \dots, p_{N-1}, 2, -\tau, \tau)^T,\\ \eta &= (-(\tau + 2q)p_0, \dots, -(\tau + 2q)p_{N-1}, 0, \tau(\tau + 2q), \tau(\tau + 2q))^T.\end{aligned} \tag{29}$$

Every entry in (27)–(28) agrees with the exact identity

$$\boxed{M = \frac{T}{2}\xi\xi^T + \frac{T}{2\tau(\tau + 2q)}\eta\eta^T \succeq 0.} \tag{30}$$

Both coefficients are positive for $q \ge 0$, $\tau > 0$. This verifies the certificate from the actual accumulator recurrence for every horizon, including the convex endpoint.

Completion of the proof of Theorem 1. Congruence by an invertible matrix preserves positive semidefiniteness. Thus (30) proves (19), and Lemma 3 gives the normalized bound. Restoring L and the translated minimizer proves (6). For $N = 0$, returning y_0 gives the bound directly from L-smoothness and $\nabla f(y_*) = 0$. The scalar assertions were proved in Section 3. ∎

4.4 Attainment on a quadratic

Proposition 4.1. Let $N \geq 0$, $L > 0$, $0 \leq \mu < L$ and $R > 0$. Apply the unchanged method to

$$f(u,v) = \frac{\mu}{2}u^2 + \frac{L}{2}v^2, \qquad y_0 = (0,R), \qquad y_* = (0,0). \tag{31}$$

Then

$$\|\nabla f(y_N)\| = L\tau_N(\mu/L)R. \tag{32}$$

Consequently no uniformly smaller factor can replace τ_N in the guarantee of this method.

Proof. For $N = 0$ the claim is immediate. Suppose $N \geq 1$ and normalize $L = 1$. The first coordinate remains zero, so write z_i for the second coordinate of y_i. On this trajectory $s_i = z_i$, and the transformed scalar data are $x_i = 0$, $g_i = z_i$. Put

$$c = \rho_N = \frac{q+\tau}{1+q} = 1 - p_{N-1}, \qquad Q = \sum_{i<N} \Delta p_i z_i.$$

The actual accumulator input in (4) is $w_i = q(z_i - R) - (1+q)z_i = -z_i - qR$. Hence its final first accumulator is $-Q - qp_{N-1}R$. Since $(1+q)T = 1/c$, the output equation gives

$$z_N = R - \frac{Q + qp_{N-1}R}{c}, \qquad Q + cz_N = (c - qp_{N-1})R = \tau R. \tag{33}$$

Evaluate the lifted quadratic form at the actual scalar trajectory and the minimizing auxiliary value $-z_N/\tau$. After the congruence in (30), its argument is

$$\widehat{z} = \begin{pmatrix} S^{-T}z \\ R \\ -z_N/\tau \end{pmatrix}, \qquad z = (z_0, \ldots, z_N)^T.$$

The first N coordinates of $S^{-T}z$ are $z_i - z_{i+1}$, and its last coordinate is z_N. Telescoping and (33) therefore yield

$$\xi^T\widehat{z} = Q + cz_N - \tau R = 0, \qquad \eta^T\widehat{z} = -(\tau + 2q)(Q + cz_N - \tau R) = 0.$$

Thus both Gram factors vanish on the actual trajectory. Moreover, $\operatorname{tr}(g^T\Lambda x) = 0$ because $x_i = 0$ at every index. The exact quadratic-form identity in Lemma 3 now gives

$$0 = \frac{\tau}{2}R^2 - \frac{1}{2\tau}z_N^2,$$

so $|z_N| = \tau R$. Restoring L proves (32). The objective (31) has the stated lower and upper curvature moduli; when $\mu = 0$, choosing the minimizer $(0,0)$ is still valid. ∎

Attainment concerns this particular method. A competing method need not have the same factor on the witness (31), so this proposition is not a minimax lower bound.

5 Comparison with specified guarantees

We compare coefficients for the same initial-distance normalization and N construction queries. The factor of our method is exact by Proposition 4.1; the comparisons do not identify the exact worst-case gradient performance of the three benchmark methods.

The convex class-inclusion guarantee for LemniAcc is $1/\Omega_N$ [3]. For the second benchmark, use the original ITEM coefficients from Taylor and Drori [2, Algorithm 1 and Theorem 2.1]:

$$\begin{aligned} A_0 &= 0, \\ A_{j+1} &= \frac{(1+\kappa)A_j + 2\big(1+\sqrt{(1+A_j)(1+\kappa A_j)}\big)}{(1-\kappa)^2}, \\ D_j &= (1+\kappa A_j)^{-1}. \end{aligned} \tag{34}$$

In particular, the denominator $(1-\kappa)^2$ is part of the original normalization. A block of a ITEM steps followed by b steps of its gradient-to-gradient dual [4] gives the routine norm coefficient $\sqrt{D_a D_b}$ when $a+b=N$. The first query of the second block is at the joining point, so the construction uses N queries. We include $a=0$ or $b=0$ by $D_0=1$.

The third benchmark comes from ITEM-f [3, Section 4.2, Theorem 3], which guarantees

$$f(x_N) - f_* \le \Upsilon_N^{-2}\big(f(x_0) - f_*\big). \tag{35}$$

The smoothness inequalities at x_0 and x_N convert this to the distance-to-gradient coefficient $1/\Upsilon_N$. This is a conversion of a function-gap guarantee, rather than an exact characterization of its gradient criterion.

Theorem 2. *For every $N \ge 1$ and $0 < \kappa \le 1/(64N^2)$,*

$$\tau_N(\kappa) < \min\left\{ \frac{1}{\Omega_N},\ \min_{\substack{a,b\ge 0\\ a+b=N}} \sqrt{D_a D_b},\ \frac{1}{\Upsilon_N} \right\}, \tag{36}$$

where a, b are integers and the coefficients are defined above.

Proof. Theorem 1 immediately gives

$$\tau_N(\kappa) < \tau_N(0) \le \tau_1(0) = \sqrt{2} - 1 < \tfrac{1}{2}. \tag{37}$$

To bound (34), use $\sqrt{1+x} \le 1 + \sqrt{x}$ twice. With $x = \sqrt{A_j}$ this gives

$$A_{j+1} \le \frac{\big((1+\sqrt{\kappa})x + 2\big)^2}{(1-\kappa)^2}, \qquad \sqrt{A_{j+1}} \le \frac{\sqrt{A_j}}{1-\sqrt{\kappa}} + \frac{2}{1-\kappa}.$$

Iteration yields

$$\sqrt{A_N} \le \frac{2N}{1-\kappa}(1-\sqrt{\kappa})^{-(N-1)}.$$

On the stated interval, Bernoulli's inequality gives $(1-\sqrt{\kappa})^{N-1} > 7/8$, and $1-\kappa \geq 63/64$. Hence

$$\sqrt{A_N} < \frac{1024}{441}N < 4N, \qquad D_N > \frac{4}{5}.$$

The sequence A_j increases, so $\sqrt{D_a D_b} \geq D_N > 4/5$ whenever $a+b=N$. Together with (37), this proves the second comparison.

For the ITEM-f conversion, we use the classical scalar-quadratic polynomial bound. Any fixed-coefficient first-order method using N gradients has output $P_N(\lambda)x_0$ on $f_\lambda(x) = \lambda x^2/2$, where P_N has degree at most N and $P_N(0) = 1$. If its uniform function-gap contraction factor is η^2, then

$$\eta \geq \max_{\lambda\in[\mu,L]} |P_N(\lambda)| \geq \frac{1}{\mathsf{T}_N((1+\kappa)/(1-\kappa))}, \tag{38}$$

where $\mathsf{T}_N(\cos\theta) = \cos(N\theta)$.

For completeness, map $[\mu, L]$ to $[-1, 1]$. The point $\lambda = 0$ maps to

$$z_0 = -\frac{1+\kappa}{1-\kappa} < -1.$$

The comparison polynomial $\mathsf{T}_N(z)/\mathsf{T}_N(z_0)$ has alternating extreme values at the $N+1$ points

$$\cos(j\pi/N), \qquad j = 0, \ldots, N.$$

A degree-at-most-N polynomial equal to one at z_0 with strictly smaller uniform norm would differ from this polynomial by a function with a root between every two consecutive extremal points and another root at z_0. That would give $N+1$ distinct roots, which is impossible. This proves (38), including $N = 1$.

Set $z = (1-\sqrt{\kappa})/(1+\sqrt{\kappa})$. The exact Chebyshev formula gives

$$\frac{1}{\mathsf{T}_N((1+\kappa)/(1-\kappa))} = \frac{2z^N}{1+z^{2N}} \geq z^N.$$

Since $z \geq 1-2\sqrt{\kappa} \geq 1-1/(4N)$, another use of Bernoulli's inequality gives $z^N \geq 3/4$. The printed ITEM-f method is a fixed-coefficient N-gradient construction, and (35) is uniform over the class, including these scalar quadratics. Therefore $1/\Upsilon_N \geq 3/4$. Equation (37) proves the final comparison. ∎

The interval in Theorem 2 is deliberately conservative. Its proof compares the three displayed upper coefficients and does not show that the candidate is better than every available strongly convex method. In particular, the polynomial argument lower-bounds a uniform function-gap contraction before smoothness conversion; it is not a lower bound of 3/4 for the distance-to-gradient design problem. The theorem also does not assert a positive-curvature asymptotic linear rate. The scalar factor (2) is the finite-horizon guarantee established here.

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